Anirudh Yadav Boosa

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Research Interests

Human-computer interaction and human-AI interaction; accessibility and assistive technology for blind and low-vision users; learning from what users say and do to build AI systems that adapt to people.

Education

University of North Texas Expected May 2027
B.S. in Computer Science; sole-author senior thesis advised by Dr. Kirill Morozov *Denton, TX*
Coursework: Artificial Intelligence, Data Mining, Algorithms, Software Engineering, Database Systems, Technical Writing.

State University of New York at Buffalo Jan 2023 – Dec 2024
Computer Science coursework; transferred to the University of North Texas *Buffalo, NY*

Manuscript Under Review and Poster

Journal manuscript (under review)
R. Ghoneim, W. Aljedaani, **A. Y. Boosa**, and S. Ludi. "Evaluation of Blind and Visual Impaired Android Applications." *Universal Access in the Information Society* (Springer). Under review, 2026.

Poster
R. Ghoneim, R. Sharma, A. A. Islam, S. V. Pentyala, **A. Y. Boosa**, and A.-R. Glee; supervised by S. Ludi. "What is Accessibility in Computing?" Poster, CRA UR2PhD program, University of North Texas, 2025. Accepted to the poster session of the Grace Hopper Celebration 2026 (conference canceled by its organizers).

Research Experience

Undergraduate Research Assistant, DiscoverABILITY Lab May 2025 – Jan 2026
University of North Texas. PI: Dr. Stephanie Ludi; day-to-day mentor: Dr. Rana Ghoneim *Denton, TX*

- Co-authored an empirical study of 26 Android apps built for blind and visually impaired users, pairing automated accessibility evaluation with 8,286 Google Play reviews of those apps (manuscript listed above).
- Ran Google Accessibility Scanner on onboarding, interactive and feedback screens and categorized 1,810 violations into 12 dimensions (touch target, contrast, metadata, focus order and others).
- Built Python pipelines (pandas, scikit-learn, NLTK; TF-IDF features, Random Forest) that clean, vectorize and classify the reviews into multi-label accessibility failure categories.
- Related the scanner results to what reviewers reported in their own words, to derive design recommendations for screen-reader (TalkBack) users; helped write and format the manuscript.

UR2PhD Research Scholar Jun 2025 – Jul 2025
Computing Research Association (CRA) and University of North Texas. Mentor: Dr. Stephanie Ludi *Denton, TX*

- Selected for the CRA UR2PhD program (funded, $8,000): a summer research project on accessible mobile systems with a faculty mentor and a PhD student mentor, plus the research training course (research methods, scientific writing, research ethics).
- Co-authored the poster "What is Accessibility in Computing?" with Dr. Rana Ghoneim and four other UR2PhD scholars.

Independent Research: AI triage of accessibility bug reports Nov 2026 – Present
Exploratory. Open source; code and documentation written with AI assistance, as each README states *Denton, TX*

- **TalkBack Triage Pilot** (github.com/anirudh763/talkback-triage-pilot; live app: anirudh763.github.io/talkback-triage-pilot). Proposed to Dr. Wajdi Aljedaani as a follow-up to the lab study: do TalkBack reproduction steps and the app version in a bug report change how a large language model (LLM) triages it? Built a web app that tests this on 75 real Android accessibility issues from two open-source apps.
- Exploratory result, one LLM run: of 32 reports with TalkBack steps, removing only the steps switched the model to "needs more evidence" on 12 and the other way on 1 (exact McNemar $p = 0.003$); of 49 reports with version or device details, removing them switched 2 and none the other way ($p = 0.50$).

- Manipulation check: the model named the steps as missing on 25 of the 32 reports once they were removed, against 1 of 32 when present. Each shortened copy is checked to differ from the original only in the removed detail, and every removed line is published.
- No human labels yet. The app lets raters label the same reports without seeing the model's answers, and the labeling protocol (independent raters, shared rubric, agreement by Cohen's kappa) and a rule-based baseline are written and public.
- **a11y-issue-warehouse** (github.com/anirudh763/a11y-issue-warehouse). 2,643 issues from 15 open-source Android apps (1,187 accessibility reports and a date-matched comparison sample) in a tested SQLite warehouse with 19 data-quality checks; one check caught 180 keyword-search results that were not about accessibility.
- A leave-one-repository-out text classifier (TF-IDF, logistic regression) reaches F1 0.756 on unseen repositories with the defining keywords masked (95% bootstrap interval 0.736 to 0.776). Labels are not hand-verified, and the README lists the limits.

Senior Thesis Researcher Aug 2026 – Present
University of North Texas. Advisor: Dr. Kirill Morozov *Denton, TX*

- Sole-author thesis on control-aware attacks and defenses for drones that run ArduPilot and communicate over MAVLink: developing command validation and anomaly detection, working independently with advisor feedback.

Projects

Network IoT Detection, Senior Capstone (team of 4) Aug 2026 – May 2027
CSCE 4901/4902, University of North Texas *Denton, TX*

- Smart-home IoT security monitor that alerts blind and low-vision users through screen-reader output and audio sonification; I am responsible for the non-functional requirements (data protection, access control, uptime, network recovery).

Other open-source software (github.com/anirudh763; code written with AI assistance, as each README states): flightseal, a tamper-evident log for drone flight records (C++17); mavframe, a MAVLink v2 parser that uses no heap memory, fuzzed and checked against pymavlink (C99); leaseq, a crash-safe job queue (Python, PostgreSQL, FastAPI).

Contribution to Apple open source: fix merged into apple/pkl-swift (PR #138, Apr 2026). HTTP settings merged from a PklProject's evaluator settings were dropping custom TLS CA certificates; the fix keeps them. Reviewed, approved and merged by an Apple maintainer.

Awards and Selections

STARS AI Scholar, STARS Computing Core, 2026–27 cohort (K-12 AI education projects) Nov 2026
Poster accepted, Grace Hopper Celebration 2026 (conference canceled by its organizers) Sep 2026
CRA UR2PhD Summer Research Award ($8,000) and research training course certificate Summer 2025

Programs and Community

AccessComputing, member; national alliance based at the University of Washington 2025 – Present
Applied Research and Engineering Practice (AREP), UNT Libraries; 10-week team research cohort Fall 2026
Student Leader and Peer Mentor, University at Buffalo Jan 2024 – Dec 2024
Member, UNT Computer Science Club and Chess Club; CCBA Mentorship Program

Employment

Front Desk Assistant, UNT Housing and Residence Life, Area 3 (four residence halls) May 2025 – Present
Student Employee, Campus Dining and Shops, University at Buffalo Aug 2023 – Sep 2024

Skills

HCI and accessibility: accessibility evaluation (Google Accessibility Scanner, TalkBack, WCAG), qualitative and quantitative analysis of user reviews and bug reports, scientific writing
Data and machine learning: Python (pandas, NumPy, scikit-learn, NLTK, Matplotlib), text classification, SQL (SQLite, PostgreSQL), generative AI (LLM) prompting and evaluation
Programming and tools: Python, C++, C, Java, JavaScript, HTML/CSS, Bash, Git, GitHub, Linux, LaTeX
Spoken languages: English, Hindi, Telugu